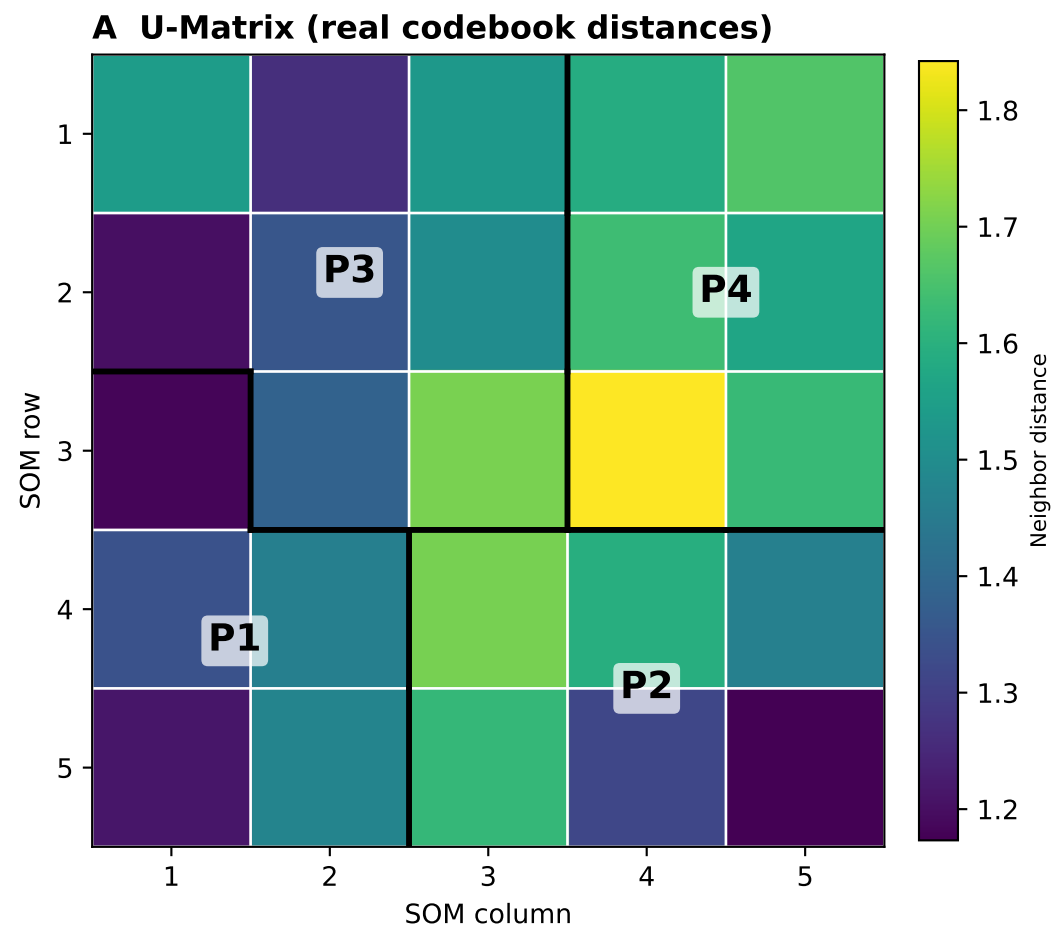
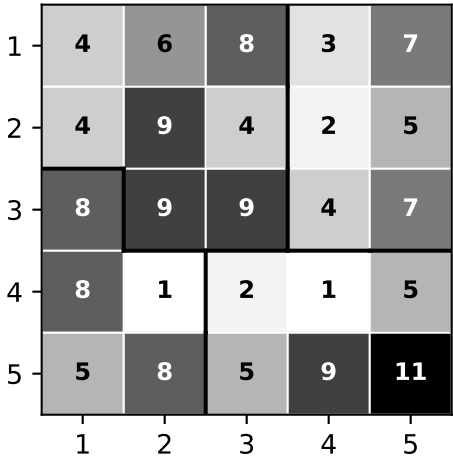


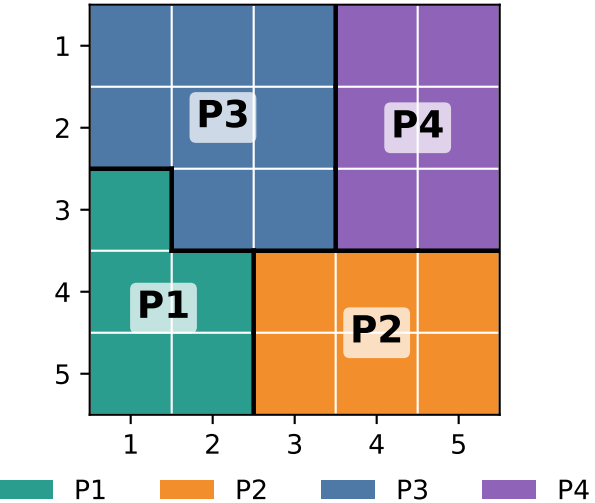
Stage 5 — Real SOM diagnostics for Pará municipalities  
Selected 5×5 SOM, seed 5; all values derived from frozen model outputs



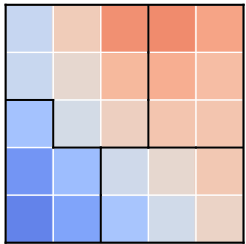
**B BMU hits: municipalities per neuron (n=144)**



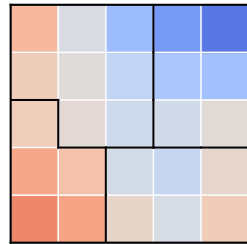
**C Four macroprofiles on the actual SOM lattice**



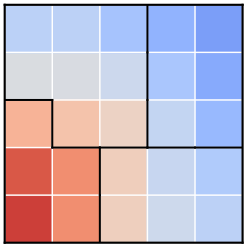
1. Rural female share



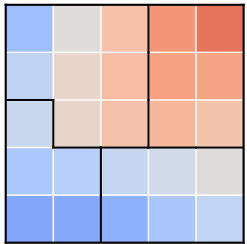
2. Female literacy 15+



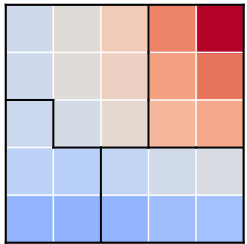
3. Household per-capita income



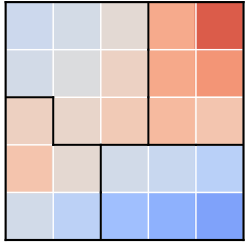
4. Age ILR1



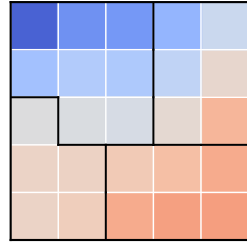
5. Age ILR2



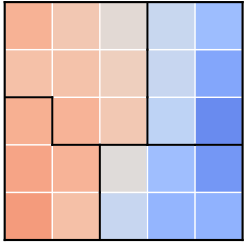
6. Age ILR3



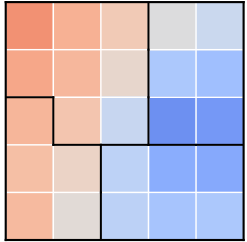
7. Race/color ILR1



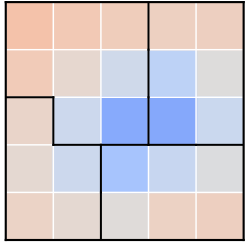
8. Race/color ILR2



9. Race/color ILR3



10. Race/color ILR4



D Component planes: codebook values in the standardized training space. Black lines mark macroprofile boundaries.